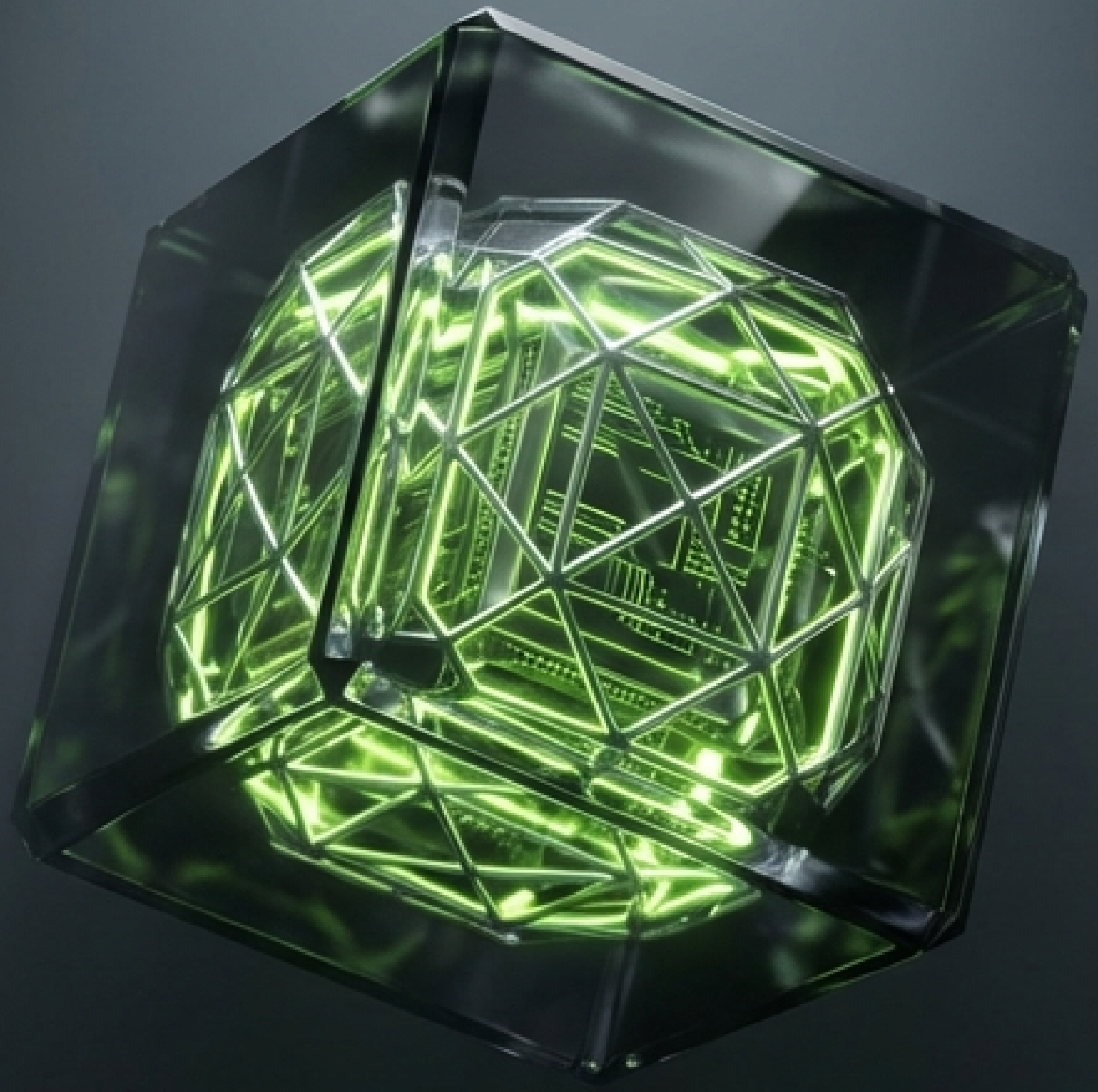


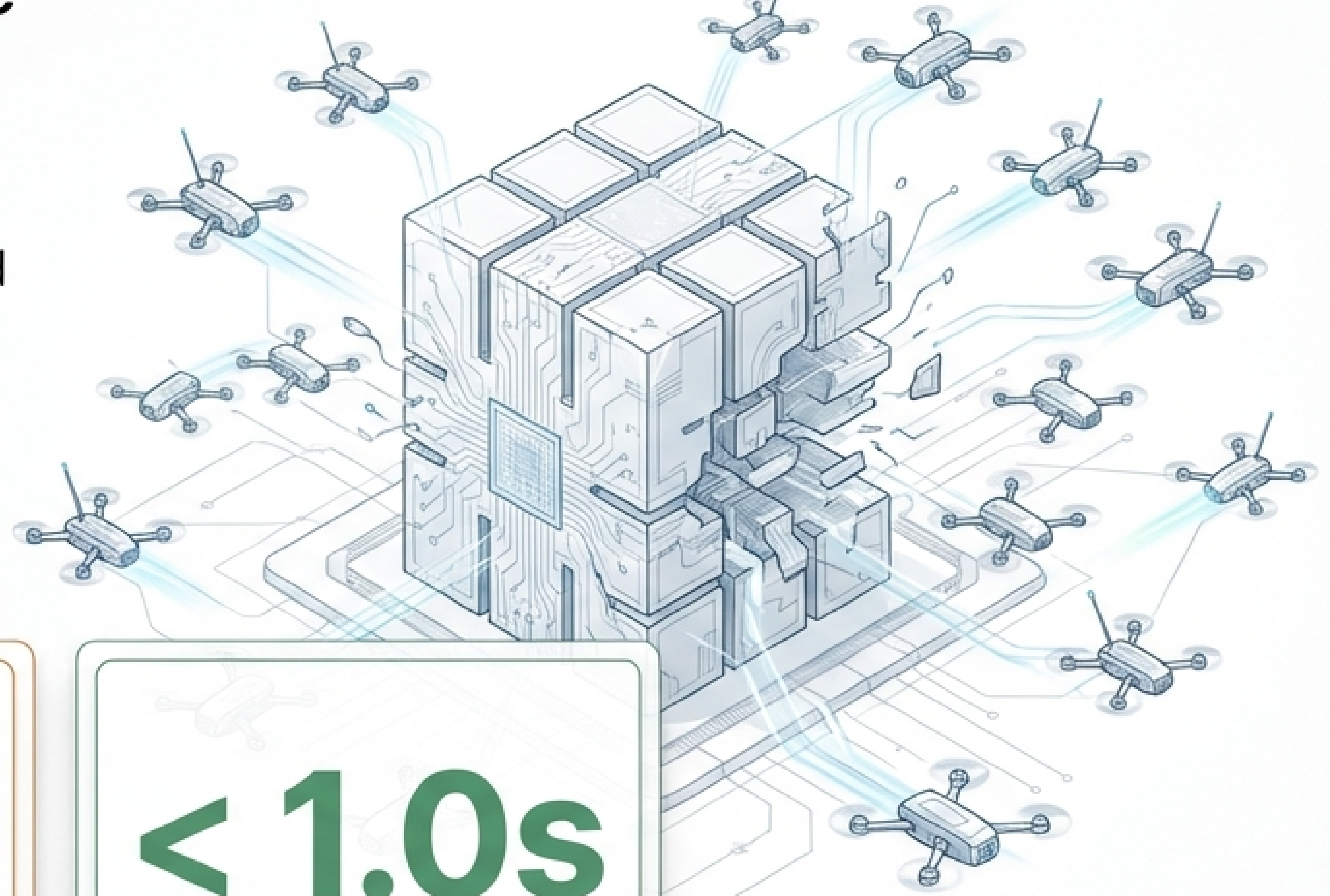
NVIDIA Model Intelligence Report 2026 Catalogue

ANALYSIS OF MODELFORGE V3.02 OPTIMISED WORKFLOWS

FEBRUARY 2026 | TECHNICAL BRIEFING

The Shift to Precision Intelligence

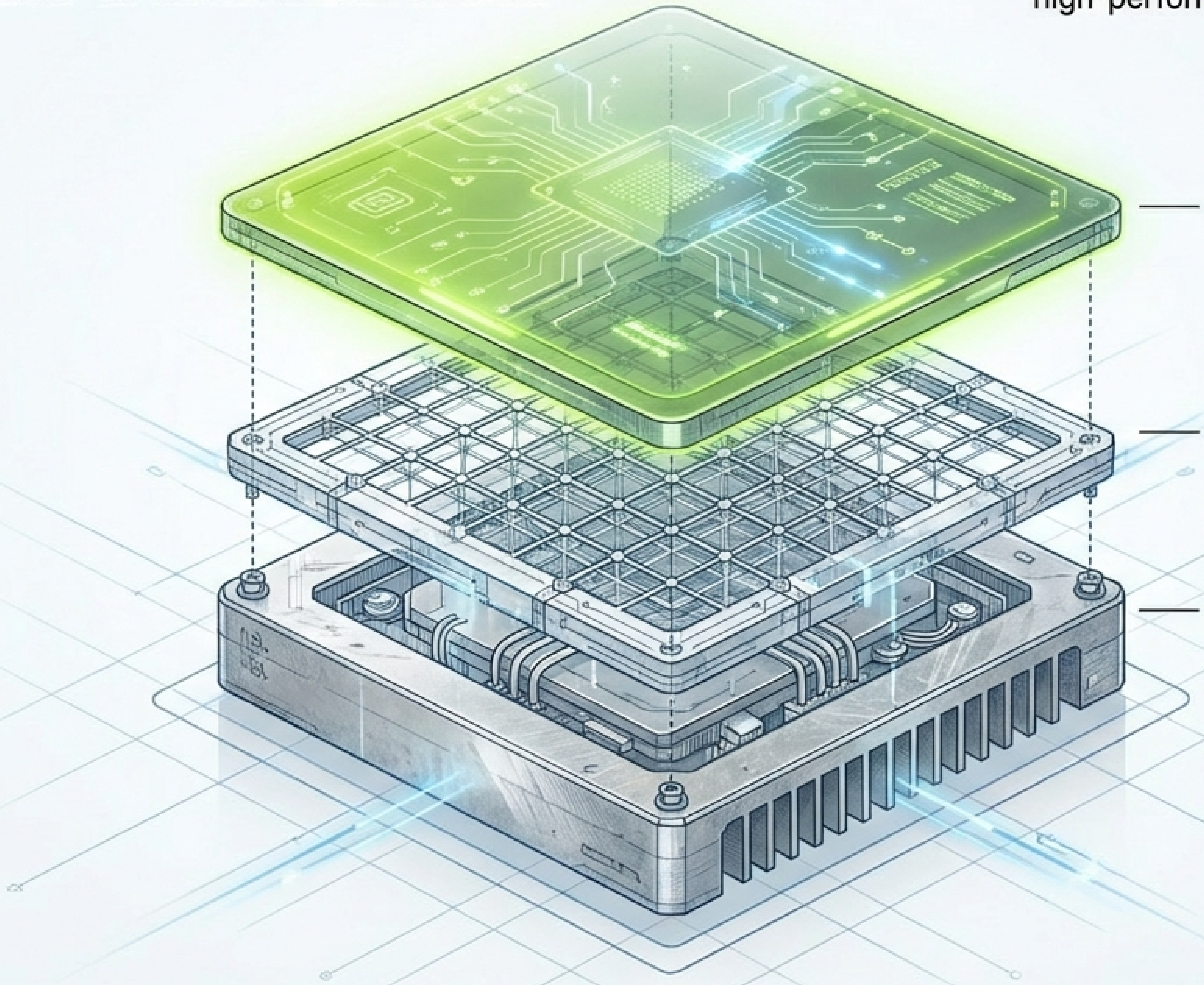
The era of the monolithic model is over. The 2026 landscape is defined by a fleet of hyper-optimised agents, leveraging Mixture-of-Experts (MoE) and Hybrid Mamba architectures to deliver billion-parameter reasoning at sub-second latencies.



3.3x	MoE & Mamba	< 1.0s
Throughput Increase	**Architecture Evolution**	**Performance Standard**
Via TensorRT-LLM & Blackwell kernels vs open-weights.	From Monolithic to Maverick & Nemotron Hybrid.	Billion-parameter reasoning latency.

The Engine Room: Infrastructure & Microservices

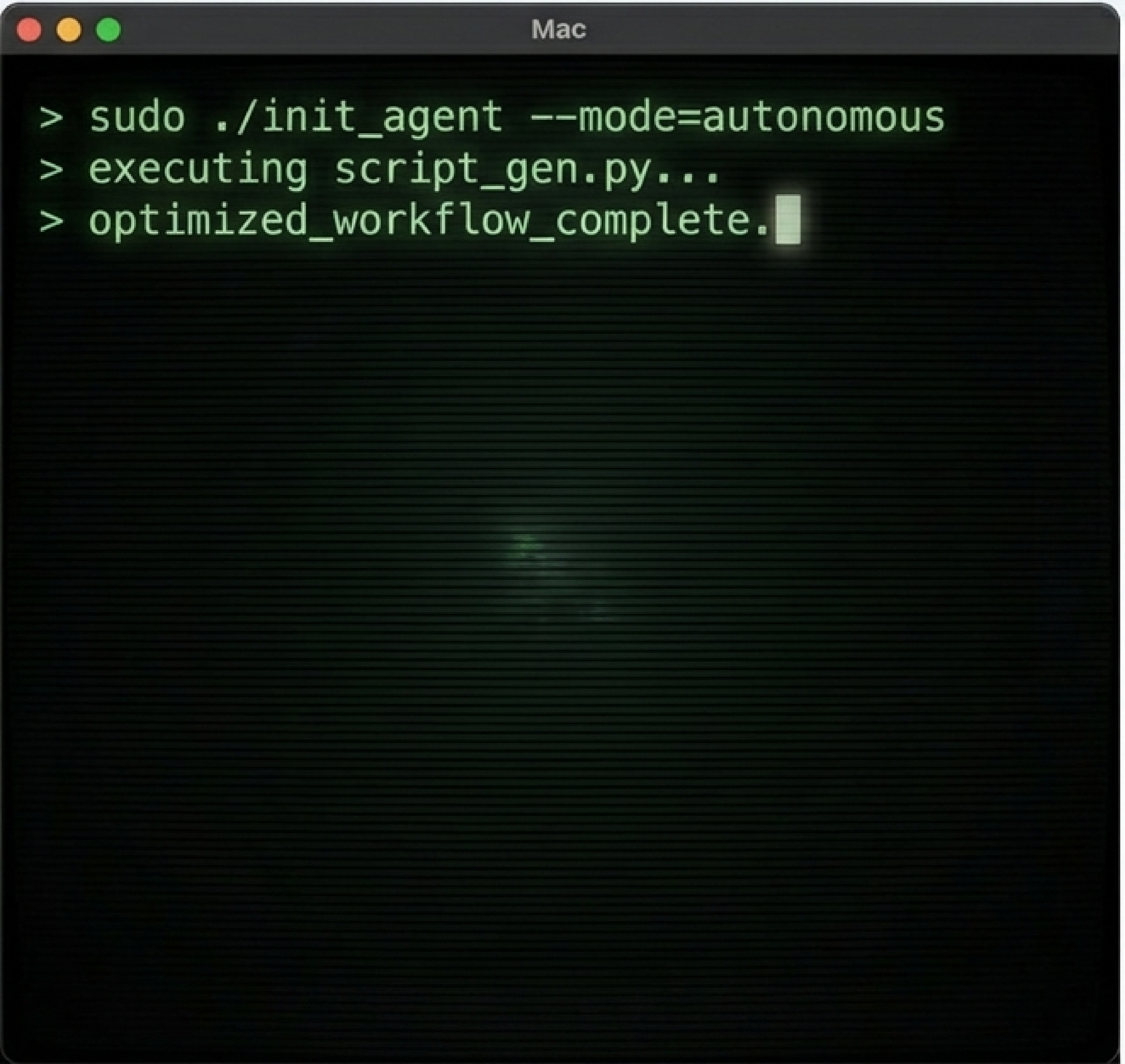
Raw weights are insufficient for enterprise scale. NVIDIA NIM provides the microservices backbone, enabling the catalogue's high-performance throughput.



The Generalist Frontier: Llama 4 Maverick

ARCHITECTURE	400B Parameters (17B Active MoE)
CONTEXT WINDOW	1,000,000 Tokens
REASONING SCORE	★★★★★ (High-Intelligence Flagship)
PRIMARY CAPABILITY	Nuanced abstract reasoning and high-fidelity multimodal understanding.
IDEAL MISSION	Complex research, creative writing, cross-modality reasoning.
OPERATIONAL NOTE	High resource overhead. Significant compute intensity compared to Scout.

The Developer’s Command Line: DeepSeek V3.1 Terminus



- **ARCHITECTURE** 671B (MoE)
- **CONTEXT WINDOW** 128,000 Tokens
- **REASONING SCORE** ★★★★★ (Agentic CLI)
- **PRIMARY CAPABILITY** Fine-tuned for terminal stability, precise tool use, and zero-shot CLI proficiency.
- **IDEAL MISSION** Automated DevOps, script generation, terminal-agent workflows.
- **CONSTRAINT** 128K context limit.

The Efficiency King: Nemotron-3 Nano

ARCHITECTURE:
30B (3.2B Active)
Hybrid Mamba

CONTEXT WINDOW:
1M Tokens

REASONING SCORE:
★★★★
(Fast Agentic Loops)

KEY METRIC:
Linear Scaling
& High
Tokens-Per-
Second (TPS)

Breaking the transformer bottleneck.
Ideal for multi-agent swarms and cost-critical loops.

Note: Requires thinking budget configuration to control verbosity.

Vision-Language-Action: Cosmos Reason2

ARCHITECTURE:
8B VLA (Vision-Language-Action)

CONTEXT WINDOW:
256K Tokens

SPECIALTY SCORE:
★★★★★
(Physics & Spatio-Temporal)

CAPABILITY:
Native spatio-temporal logic; understands video causality and 3D environments.

IDEAL MISSION:
Robotics, video analytics, simulation reasoning.

CONSTRAINT:
Low coding proficiency. Specialised for physical world.

Domain Mastery: The Coding Suite

Qwen 3 Coder 480B

The Elite Engineer

Strength: SOTA open-weights coding logic across 100+ languages.

Use Case: Precise log analysis and complex logic generation.

Minimax-M2.1

The Archive Analyst

Strength: Infinite document analysis with a 4M Context Window.

Use Case: Massive code refactoring (whole repo in context).

Note: Requires thinking budget configuration to cont

Multimodal Parsing & Retrieval

VISION: Nemotron Nano 12B VL

Task: Multimodal Doc Parsing.

131K Context.

High-fidelity visual ingestion.

AUDIO: Parakeet CTC 0.6B

Task: Real-time Transcription.

Sub-50ms latency.

Instant voice-to-text.

SEARCH: NV EmbedQA E5 v5

Task: Semantic Vector Retrieval.

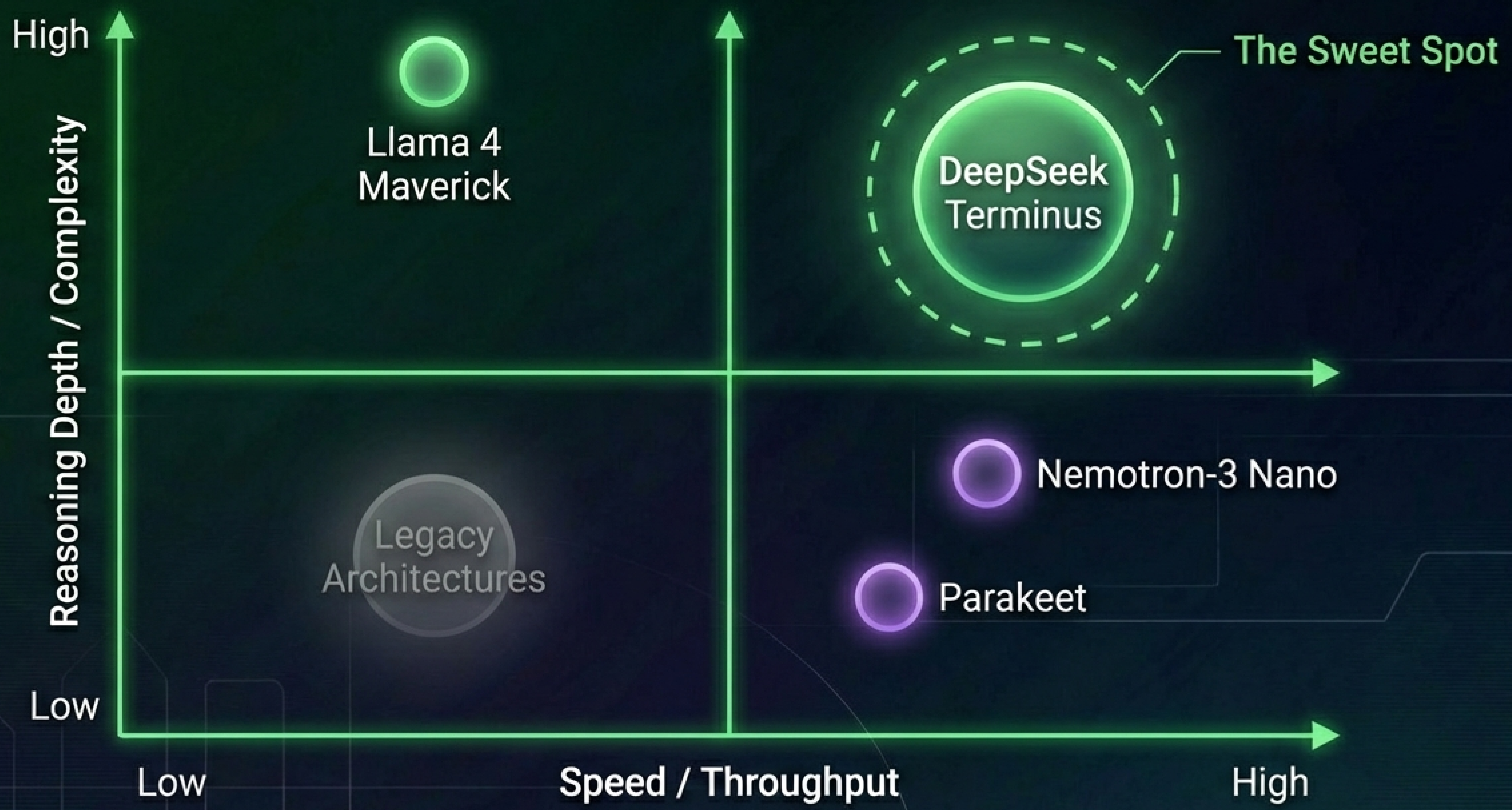
8K Dense Retrieval.

The backbone of RAG systems.

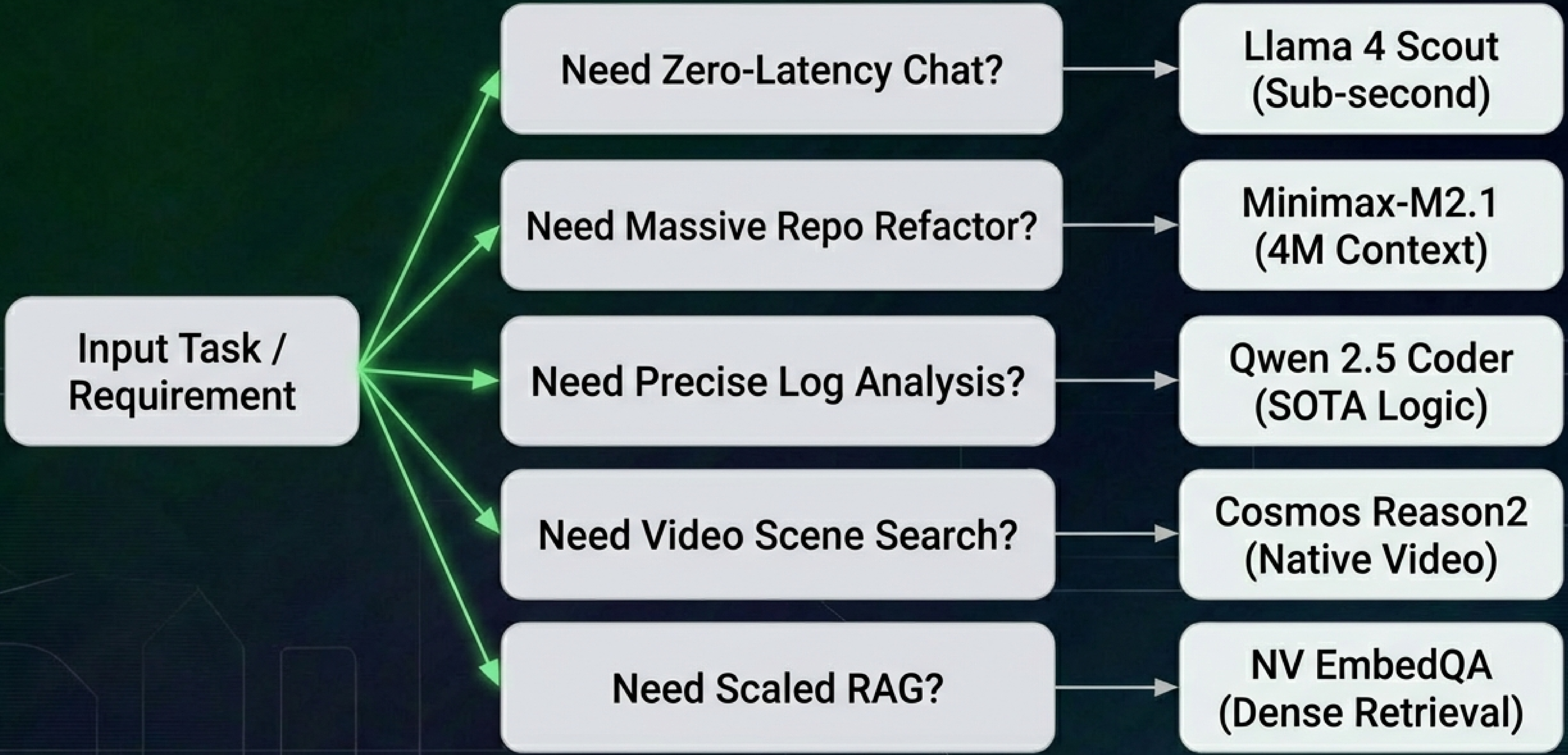
2026 Fleet Comparison Matrix

MODEL	PARAMS	CONTEXT	PRIMARY STRENGTH	REASONING
Llama 4 Maverick	400B (17B Active)	1M	High-Intelligence Flagship	★★★★★
DeepSeek V3.1 Terminus	671B (MoE)	128K	Agentic CLI	★★★★★
Nemotron Super 49B	49B (Dense)	128K	Human-Chat Alignment	★★★★
Nemotron-3 Nano 30B	30B (3.2B Active)	1M	Fast Agentic	★★★★★
Cosmos Reason2 8B	8B (VLA)	256K	Physical AI	★★★★★ (Physics)

Pivot Analysis: Capabilities vs. Throughput

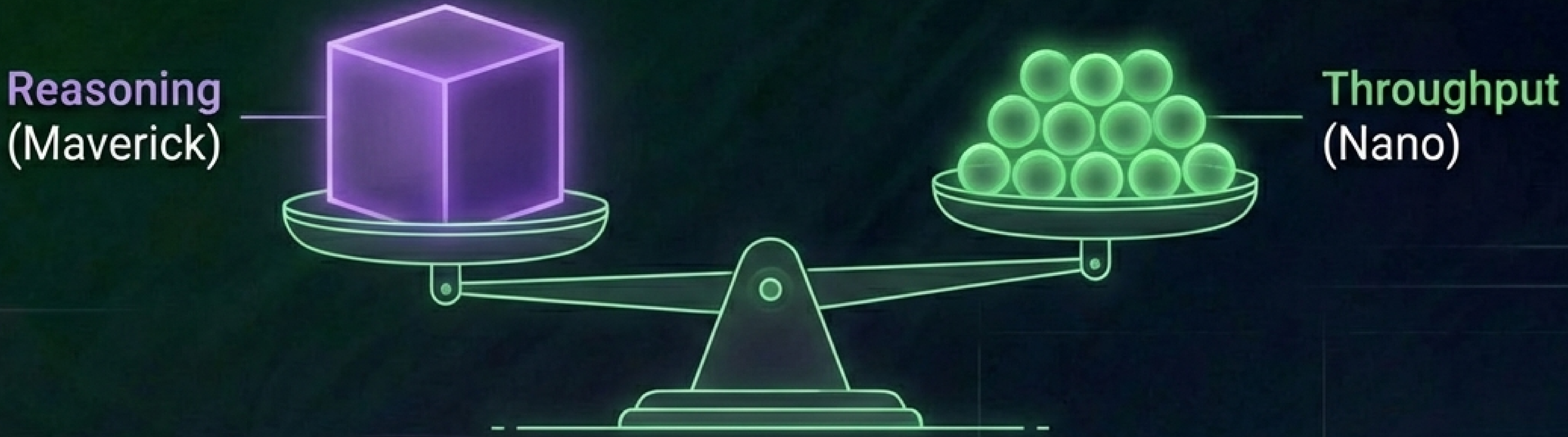


Workflow Decision Framework



Resource Strategy & Cost Management

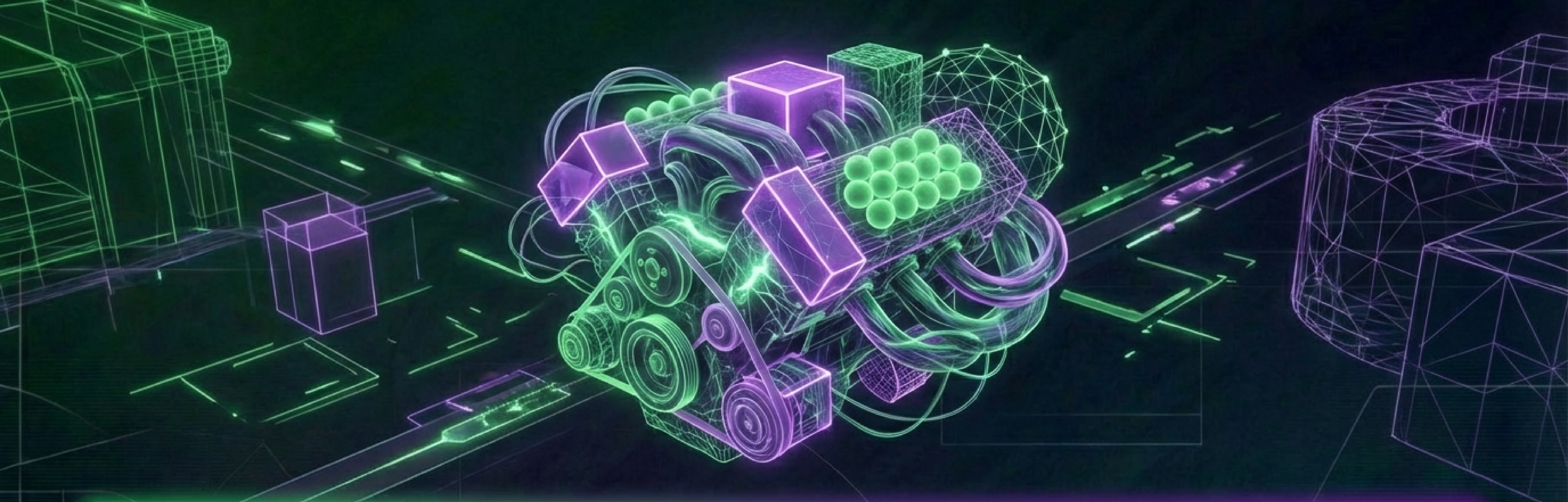
The Thinking Budget



Strategy: Utilise Nemotron-3 Nano for high-volume, low-complexity loops to offset the compute cost of invoking Llama 4 Maverick for complex reasoning tasks.

The Composite Future

The optimal workflow is no longer a single monolithic model; it is a symphony of specialised agents. Success lies in orchestrating the right tool for the right task.



Build your custom fleet within the ModelForge v3.02 workspace today.

Sources & Appendix

Sources & Appendix

PRIMARY SOURCE: nvidia_model_catalogue_presentation.md
ANALYSIS PERIOD: February 2026
PLATFORM: ModelForge v3.02
VALIDATION: Specs valid for NVIDIA-optimised kernel deployments (TensorRT-LLM).
NOTE: Performance metrics compared against standard open-weights baseline.